GEOMETRY · UNIT 6: RIGHT TRIANGLES AND TRIGONOMETRY

Accessible Ramp Engineer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design accessible ramps for school entrances. They use the rise-to-run rule, the Pythagorean Theorem, special right triangles, and trigonometric ratios to find lengths and angles, solve angle of elevation and depression problems, and design a ramp with landings.

Accessibility codes make right-triangle trigonometry meaningful. Students see that slope, ratio, and angle describe the same steepness.

Standards Addressed

- G-SRT.6 Trigonometric ratios from similar right triangles
- G-SRT.8 Use trig ratios and the Pythagorean Theorem to solve right triangles
- G-SRT.7 Sine and cosine of complementary angles

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: measure a ramp or stairs at school. Part 1 Pythagorean Theorem and the rule.
Day 2	Part 2 trig ratios and elevation.
Day 3	Part 3 design.

Materials

- Student handout
- Scientific calculators (degree mode)
- Measuring tape and level (optional field check)
- Graph paper

How the Three Versions Differ

Level	What changes
CP2	Rise-to-run rule, Pythagorean Theorem, trig ratios with a calculator, and one elevation problem.
CP1	Inverse trig, landings and switchbacks, special right triangles, and angles of depression.
Honors	Exact values, a two-observation height problem, and a design that minimizes footprint.

Watch For (Common Misconceptions)

- Calculator in radian mode.
- Using the ramp length instead of the run in the 1 : 12 rule.
- Confusing angles of elevation and depression.

Extension Ideas

- Survey ramps at school and check them against the rule.
- Research the history of accessibility laws.
- Build a scale model.

Answer Key – CP2

Part 1: Rise, Run & Length

#	Task	Answer
1	The front entrance is 30 inches above the sidewalk. What is the shortest run allowed? Give your answer in inches and feet.	30 × 12 = 360 inches = 30 feet
2	Use the Pythagorean Theorem to find the length of the ramp surface for that run.	$\sqrt{(360^2 + 30^2)} = \sqrt{130{,}500} \approx 361.2$ inches (about 30.1 ft)
3	Builders check square corners with a 6-8-10 triangle. Show that 6, 8, and 10 form a right triangle.	$6^2 + 8^2 = 36 + 64 = 100 = 10^2$

Part 2: Angles & Trig

#	Task	Answer
4	Find tan θ for the ramp (rise ÷ run). Then use $\tan^{-1}$ to find the ramp's angle θ.	$\tan\theta = 30 \div 360 \approx 0.0833$; $\theta = \tan^{-1}(0.0833) \approx 4.76°$
5	A 20-foot ladder leans against the gym at a 75° angle with the ground. How high does it reach? How far is its base from the wall?	Height = 20 sin 75° ≈ 19.32 ft; base = 20 cos 75° ≈ 5.18 ft
6	From 50 feet away, a student sees the top of the gym at an angle of elevation of 32°. Her eyes are 5 feet above the ground. How tall is the gym?	50 tan 32° ≈ 31.24; 31.24 + 5 ≈ 36.24 ft

Part 3: Design the Side Ramp

#	Task	Answer
7	Find the shortest allowed run and the ramp length.	Run 21 × 12 = 252 in = 21 ft; length $\sqrt{(252^2 + 21^2)} \approx 252.9$ in
8	Find the ramp angle.	$\tan^{-1}(21 \div 252) \approx 4.76°$ (the same angle)
9	Draw a labeled scale drawing of your ramp (for example, 1 grid square = 1 foot).	Answers vary.

Answer Key – CP1

Part 1: Rise, Run & Length

#	Task	Answer
1	For the 30-inch entrance, find the shortest run, the ramp length, and the ramp angle.	Run 360 in (30 ft); length ≈ 361.2 in; $\theta \approx 4.76°$
2	A contractor proposes a ramp with a 24-foot run for the 30-inch entrance. Does it meet the rule? Find its angle.	24 ft = 288 in; $30 \div 288 \approx 0.104 > 1 \div 12 \approx 0.083$, so it is too steep. Angle $\tan^{-1}(30 \div 288) \approx 5.95°$.
3	A 6-foot shelf needs a 45° diagonal brace from the top corner to the floor. Find the brace length exactly and as a decimal.	$6\sqrt{2} \approx 8.49$ ft
4	A lift's support strut makes a 30-60-90 triangle with a 10-foot hypotenuse. Find both legs exactly.	Short leg 5 ft, long leg $5\sqrt{3} \approx 8.66$ ft

Part 2: Elevation & Depression

#	Task	Answer
5	From the roof (40 ft high), a worker sees a delivery truck at an angle of depression of 25°. How far is the truck from the building?	$40 \div \tan 25° \approx 85.78$ ft
6	From 50 ft away, eyes 5 ft high, the angle of elevation to the top of the gym is 32°. Find the gym's height.	≈ 36.24 ft
7	A ramp rises 30 inches over a 40-foot run. Find its angle and its length in feet.	40 ft = 480 in: $\theta = \tan^{-1}(30 \div 480) \approx 3.58°$; length $\sqrt{(480^2 + 30^2)} \approx 480.9$ in ≈ 40.08 ft

Part 3: Design the Gym Ramp

#	Task	Answer
8	Plan the ramp in two equal sections. Find the rise and the shortest run of each section.	22.5 in of rise each; run $22.5 \times 12 = 270$ in = 22.5 ft each
9	Find each section's length and the total run including a 5-foot landing.	Each length $\sqrt{(270^2 + 22.5^2)} \approx 270.9$ in ≈ 22.58 ft; total run $22.5 + 5 + 22.5 = 50$ ft
10	Sketch a switchback design and give its footprint.	Answers vary.

Answer Key – Honors

Part 1: Exact Steepness

#	Task	Answer
1	Show that every ramp meeting the rule exactly has the same angle, $\theta = \tan^{-1}(1 \div 12)$. Find $\sin\theta$ and $\cos\theta$ exactly.	The rise-to-run ratio is always 1 : 12, so $\tan\theta = 1 \div 12$ for all of them. Hypotenuse $\sqrt{145}$: $\sin\theta = 1 \div \sqrt{145} = \sqrt{145} \div 145$, $\cos\theta = 12 \div \sqrt{145} = 12\sqrt{145} \div 145$.
2	Prove that $\sin\theta = \cos(90° - \theta)$ using the ramp triangle.	In a right triangle, the side opposite θ is adjacent to the other acute angle $(90° - \theta)$, so $\sin\theta$ = opposite ÷ hypotenuse = $\cos(90° - \theta)$.
3	A ramp must rise 30 inches but the space allows only 25 feet of run. What is the smallest number of inches the entrance must be lowered (or what…	25 ft = 300 in allows at most 25 in of rise; the rise must drop by 5 in (or add a landing and a second section).

Part 2: Two Observations

#	Task	Answer
4	From point A the angle of elevation to the top of the gym is 30°. From point B, 40 ft closer, it is 45°. Find the height exactly and as a decimal…	$h = 40\tan30°\tan45° \div (\tan45° - \tan30°) = 20(\sqrt{3} + 1) \approx 54.64$ ft
5	From the roof (40 ft high), the angles of depression to two cars in a line are 25° and 40°. How far apart are the cars?	$40 \div \tan25° \approx 85.78$ ft and $40 \div \tan40° \approx 47.67$ ft; about 38.11 ft apart
6	Write a general formula for the height h if the angles are α and β ($\beta > \alpha$) and the points are d apart.	$h = d\tan\alpha\tan\beta \div (\tan\beta - \tan\alpha)$

Part 3: Optimize the Gym Ramp

#	Task	Answer
7	Design a switchback ramp that fits in the 30-ft depth. How many sections and landings do you need?	Example: two 22.5-ft sections side by side with one 5-ft turning landing (rise 22.5 in each).
8	Find the total ramp length and the total footprint area.	Answers vary.
9	Justify that your design meets every rule and explain one trade-off.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Ramp design meets the rise-to-run rule and landing rule, with correct right-triangle calculations. Goes beyond the minimum with a creative, realistic choice.	Ramp design meets the rise-to-run rule and landing rule, with correct right-triangle calculations.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes a scale drawing, run, ramp length, angle, landings, and a check against the rule. Polished, organized, and visually engaging.	Plan includes a scale drawing, run, ramp length, angle, landings, and a check against the rule. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.